

SoulScope Whole-Scan Pattern Registry v0.1

Status: Canonical Scientific Implementation Companion — Final Architecture, Calibration Pending

Version: 0.1

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Governing authority: SoulScope Canon v1.3

Upstream authorities: Constellation Dimension Registry v0.1; Inference Rule Registry v0.1; Constellation State Registry v0.1; Cross-Constellation Interaction Registry v0.1

Downstream authority: Narrative Registry v0.1

Production modality: Voice

NORTH STAR

A Pattern summarizes the strongest whole-scan structure already established by the evidence. It does not create new meaning.

A SoulScope Pattern describes the organization of a completed scan.

It does not describe the identity, personality, diagnosis, emotional essence, or permanent nature of the person.

The Pattern layer is optional.

No Pattern is a valid result.

1. Purpose

The Whole-Scan Pattern Registry defines how SoulScope may produce a compact summary of the completed semantic result after all lower-level scientific reasoning is finished.

The Pattern Engine operates only after:

Measurements → Evidence → Dimensions → Constellation Geometry → States / Blends / Unresolved → Interaction

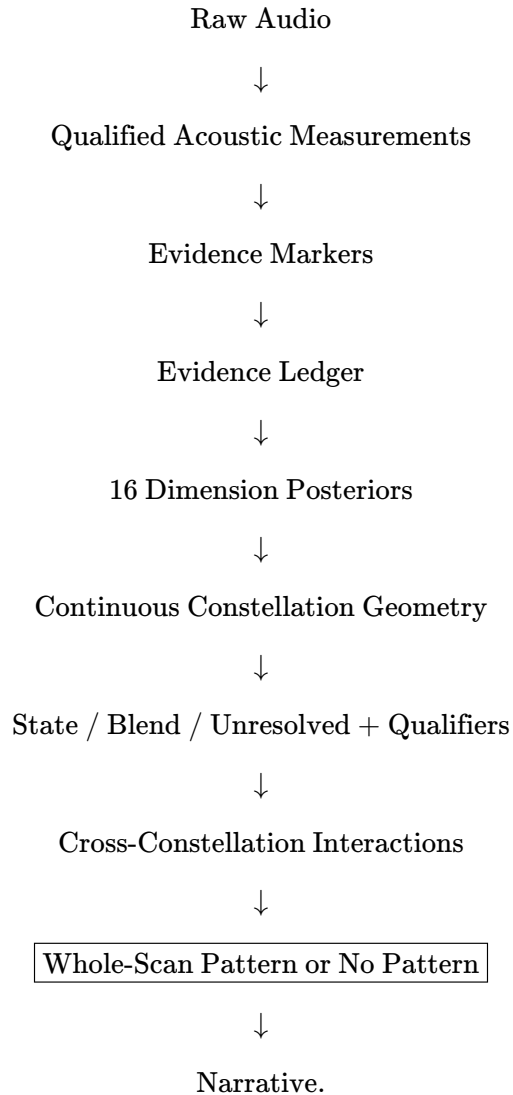
Only then may SoulScope evaluate:

Whole-Scan Pattern

The Pattern Engine SHALL NOT perform new acoustic, evidentiary, dimensional, State, or Interaction inference.

2. Canonical semantic sequence

The authoritative semantic branch is:



At no point may a downstream layer manufacture information absent upstream.

3. Pattern definition

A Pattern is:

A compact whole-scan motif representing a sufficiently supported relationship among completed Constellation structures.

A Pattern is not:

- a personality type;
- an archetype;
- a diagnosis;
- an emotional category;
- a behavioral identity;
- a clinical classification;
- a coping style;
- an attachment style;
- a spiritual state;
- a hidden truth;
- a prediction of future behavior.

A Pattern is bounded to the scan and its valid reference context.

4. Pattern is optional

SoulScope SHALL NOT require every valid scan to receive a named Pattern.

A valid completed scan may produce:

NO_PATTERN_PUBLISHED

while still containing useful:

- Measurements;
- Evidence;
- Dimensions;
- Constellation geometry;
- States;
- Qualifiers;
- Interactions.

Pattern abstention does not mean scan failure.

5. Anti-combinatorial principle

SoulScope SHALL NOT create a unique Pattern for every combination of:

- four Constellations;
- sixteen Dimensions;

- State regions;
- qualifiers;
- Interactions;
- prompt transitions;
- uncertainty states;
- longitudinal differences.

Instead, the canonical grammar is:

Compact Pattern + at most one Pattern Modifier

with the underlying Constellation and Interaction structure preserved separately.

A configuration does not deserve a new Pattern name merely because it is mathematically possible.

6. Pattern authority

The Pattern Engine receives completed scientific results.

It does not create them.

Hard invariant

Pattern authority < upstream scientific authority

A Pattern SHALL NOT:

- modify a Dimension;
 - resolve an unresolved Dimension;
 - select a different State;
 - replace a Boundary Blend;
 - invent an Interaction;
 - infer unavailable protocol constructs;
 - change Evidence;
 - rescore audio;
 - change confidence upstream.
-

7. Pattern input contract

The Pattern Engine receives:

```
PatternInput

scan_id
protocol_version

constellations:
  COG
  REG
  CAP
  EXP

dimension_posteriors[]

state_results[]
boundary_blends[]
qualifiers[]

published_interactions[]
interaction_cautions[]

cross_domain_contradictions[]

prompt_context_results[]
temporal_changes[]

overall_quality
overall_coverage

registry_versions[]
```

Research-only upstream constructs SHALL NOT silently enter a production Pattern.

8. Pattern uses continuous geometry

State labels alone are insufficient.

The Pattern Engine SHALL retain access to:

- Dimension posterior means;
- posterior intervals;
- confidence;
- Evidence coverage;
- baseline trust;
- contradiction;

- coherence;
- momentum;
- resolution status.

Therefore:

$$Pattern = f(\text{geometry, States, Qualifiers, Interactions, uncertainty}).$$

It is not:

$$Pattern = f(\text{four State names}).$$

9. Pattern object

Every registered Pattern SHALL contain:

```

pattern_id
pattern_version

canonical_internal_name

scientific_class
scientific_status

required_constellations[]
required_dimensions[]
required_interactions[]

optional_dimensions[]
optional_interactions[]

required_structural_conditions[]

contradicting_conditions[]
hard_exclusions[]

minimum_coverage_policy
minimum_confidence_policy
minimum_pattern_fit_policy

adjacent_patterns[]

permitted_modifiers[]

prohibited_meanings[]

```

user_language_key

activation_status

validation_status

10. Scientific classes

P1 – STRUCTURAL_SUMMARY

Summarizes comparatively direct organization among resolved functional structures.

Examples:

- coherent maintenance;
- contextual reconfiguration.

P2 – COST_RELATIONSHIP

Summarizes the relationship between maintained output and observable acoustic/temporal cost.

Example:

- maintained with cost.

P3 – CROSS_DOMAIN_CONFIGURATION

Summarizes a meaningful mismatch or configuration involving multiple functional domains.

Example:

- cross-domain mismatch.

P4 – TEMPORAL_PATTERN

Summarizes a meaningful change in the relationship among Constellations across prompts or compatible longitudinal scans.

Example:

- temporal relation shift.
-

11. Scientific status

Canonical statuses are:

```
ENGINEERING_PRIOR
RESEARCH_ONLY
RELIABILITY_VERIFIED
CONSTRUCT_VALIDATED
CALIBRATED
PRODUCTION_ACTIVE
DEPRECATED
```

A Pattern cannot have a scientific status stronger than its weakest required upstream construct.

$$Status(P) \leq \min_i Status(U_i)$$

where U_i are required upstream Dimensions, States, or Interactions.

12. Current activation status

The Pattern architecture itself is canonical.

The initial Pattern motifs defined in this registry are scientific candidates.

Until validated and calibrated on SoulScope data, their status is:

```
RESEARCH_ONLY
```

unless a later registry version explicitly promotes them.

No production deployment SHALL represent these candidate motifs as scientifically validated merely because their IDs exist.

13. Pattern eligibility

A Pattern candidate is eligible only when:

1. all required upstream constructs exist;
2. all required constructs are resolved;

3. all required constructs meet the active scientific-status policy;
4. required Constellation coverage is sufficient;
5. required Interactions are available where specified;
6. source Evidence independence is sufficient;
7. no hard exclusion applies;
8. active protocol is compatible;
9. Pattern model/version is compatible;
10. contradiction remains within active policy;
11. candidate fit meets the active eligibility policy.

Failure of a hard requirement makes the Pattern candidate ineligible.

14. Pattern fit

A Pattern defines a registered structural region:

$$R_P.$$

Conceptually, for completed semantic structure $\mathbf{x}$:

$$Fit_P = E_{\mathbf{x}}[\phi_P(\mathbf{x})]$$

where:

$$\phi_P(\mathbf{x}) \in [0, 1]$$

represents compatibility with the Pattern region.

The exact region model may later use:

- transparent rules;
- posterior overlap;
- calibrated density models;
- hierarchical models;
- other validated interpretable methods.

Hard boundary

Until calibration:

`pattern_fit`

means:

compatibility with the registered motif.

It does not mean:

probability that this is the person's true Pattern.

15. Candidate scores do not have to sum to one

SoulScope SHALL NOT normalize Pattern candidates merely so that one candidate wins.

The following architecture is prohibited unless future validation explicitly justifies it:

$$\sum_P P(P) = 1$$

with mandatory winner selection.

All Pattern candidates may legitimately fit poorly.

When no candidate clears publication requirements:

`NO_PATTERN_PUBLISHED`

is required.

16. Pattern confidence

Pattern confidence is separate from Pattern fit.

Conceptually:

$$C_P = f(\textit{Fit}_P, C_U, \textit{Coverage}_P, \textit{Independence}_P, \textit{Contradiction}_P, \textit{Separation}_P, \textit{Validation}_P)$$

where:

- C_U = upstream confidence;
- $\textit{Coverage}_P$ = required structural coverage;
- $\textit{Independence}_P$ = independence of supporting Evidence;
- $\textit{Contradiction}_P$ = contradictory structure;
- $\textit{Separation}_P$ = separation from viable alternatives;
- $\textit{Validation}_P$ = Pattern-model validation quality.

A Pattern may have high structural fit and only moderate confidence.

17. Evidence independence

The Pattern Engine SHALL prevent repeated use of one acoustic phenomenon from appearing as multiple independent confirmations.

For example:

F0 shift → Activation

and:

same F0 shift → Mobilization

must not automatically be counted as two fully independent sources supporting a Pattern.

Upstream Evidence-overlap information SHALL propagate into Pattern confidence.

18. Contradiction

Contradiction is first-class Pattern information.

A Pattern may be strongly supported by several structures and materially contradicted by another.

The Pattern Engine SHALL preserve:

```
supporting_structures[]  
contradicting_structures[]
```

Contradiction may:

- reduce confidence;
- widen uncertainty;
- favor an alternative;
- produce a Boundary Pattern;
- produce `NO_PATTERN_PUBLISHED`.

Contradiction SHALL NOT disappear through averaging.

19. Pattern outcomes

Canonical Pattern outcomes are:

PATTERN
PATTERN_WITH_MODIFIER
BOUNDARY_PATTERN
NO_PATTERN_PUBLISHED
SUPPRESSED
INVALID

20. PATTERN

Returned when:

- one eligible Pattern clears publication requirements;
- candidate separation is sufficient;
- confidence is sufficient;
- no hard exclusion applies.

21. PATTERN_WITH_MODIFIER

Returned when one dominant Pattern is clear and one independently resolved secondary characteristic materially improves the summary.

A Pattern may have at most:

1

Pattern-level modifier.

Additional information remains represented through Constellation qualifiers and Interactions.

22. BOUNDARY_PATTERN

Returned when two explicitly adjacent Pattern candidates:

- are both sufficiently supported;
- overlap materially;
- neither clearly dominates;
- satisfy the active blend policy.

SoulScope SHALL preserve the boundary rather than force a winner.

23. Pattern adjacency

Boundary relationships must be explicitly registered.

Two Patterns may not blend merely because their fit values are numerically similar.

If two strongly supported candidates are nonadjacent:

NO_PATTERN_PUBLISHED

is preferred.

Reason:

NONADJACENT_COMPETING_PATTERNS

24. More than two plausible Patterns

If three or more materially plausible Pattern candidates remain:

NO_PATTERN_PUBLISHED

with:

MULTIPLE_PLAUSIBLE_MOTIFS

is preferred.

SoulScope SHALL NOT concatenate three Pattern labels.

25. Pattern abstention

Canonical abstention reasons include:

INSUFFICIENT_CONSTELLATION_COVERAGE
REQUIRED_DIMENSION_UNRESOLVED
REQUIRED_INTERACTION_UNRESOLVED
PATTERN_MODEL_NOT_VALIDATED
PATTERN_FIT_TOO_LOW
LOW_PATTERN_CONFIDENCE
HIGH_CROSS_DOMAIN_CONTRADICTION
MULTIPLE_PLAUSIBLE_MOTIFS
NONADJACENT_COMPETING_PATTERNS
INSUFFICIENT_EVIDENCE_INDEPENDENCE
MODEL_OUT_OF_DISTRIBUTION
ACTIVE_HARD_CONFOUND
PROTOCOL_INCOMPATIBLE

All relevant reasons SHALL be preserved.

26. Protocol-unobservable Dimensions

Under the current three-prompt production protocol:

REG-P4 Recovery
CAP-P2 Reserve
EXP-P4 Relational Availability

are structurally unresolved.

No Pattern may infer, reconstruct, imply, or substitute for these Dimensions.

Therefore:

Pattern $\nRightarrow$ Recovery

Pattern $\nRightarrow$ Reserve

Pattern $\nRightarrow$ Relational Availability.

27. Pattern naming law

Pattern names SHALL describe measured structure.

Preferred naming form:

FUNCTIONAL_STRUCTURE

rather than:

PERSON_TYPE

Prohibited naming styles include:

- The Resilient One;
- The Overthinker;
- The Achiever;
- The Protector;
- The Empath;
- The Masker;
- The Avoidant;
- The Regulator;
- The Explorer;
- any comparable identity/archetype construction.

28. Internal name versus user language

Every Pattern maintains:

canonical_internal_name
user_language_key

The internal Pattern may be:

MAINTAINED_WITH_COST

while user-facing wording may be:

Maintained with greater observable cost

The Narrative layer determines final wording.

Changing user copy SHALL NOT change Pattern geometry or scientific meaning.

29. Initial Pattern grammar

The initial research grammar contains seven candidate motifs:

ID	Canonical Pattern
PTN-S01	COHERENT_MAINTENANCE
PTN-S02	MOBILIZED_MAINTENANCE
PTN-S03	MAINTAINED_WITH_COST
PTN-S04	CONTEXTUAL_RECONFIGURATION
PTN-S05	VARIABLE_RECONFIGURATION
PTN-S06	CROSS_DOMAIN_MISMATCH
PTN-S07	TEMPORAL_RELATION_SHIFT

These candidates begin as:

RESEARCH_ONLY

Their existence does not guarantee that all seven will survive empirical validation.

30. PTN-S01 — COHERENT_MAINTENANCE

Class: P1

Status: RESEARCH_ONLY

Canonical meaning

Several independently supported functional domains remain comparatively organized, stable, and maintained across the scan with limited cross-domain contradiction.

Candidate required structure

Strong support should generally involve at least three functionally distinct structures, with priority given to:

COG Organization and/or Focus Continuity
REG Stability
CAP Sustainability

Possible additional support:

EXP Range or deployment
validated REINFORCES interaction
validated INTEGRATES interaction

Candidate interpretation

Multiple parts of the observed vocal pattern remained mutually coherent and maintained.

Prohibited meanings

This Pattern does not mean:

healthy
regulated
resilient
optimal
emotionally stable
well-adjusted
strong person

31. Coherent Maintenance is not low Activation

COHERENT_MAINTENANCE does not require low Activation.

A scan may show:

higher Activation
+
strong Organization
+
strong Stability

+
maintained Sustainability

and still be structurally coherent.

Activation and cost must remain separately represented.

32. PTN-S02 — MOBILIZED_MAINTENANCE

Class: P1 / P2

Status: RESEARCH_ONLY

Canonical meaning

Observable mobilization increases while speech-production organization and short-interval output remain comparatively maintained.

Candidate required structure

One or more of:

REG Activation higher
CAP Mobilization higher

plus maintained support from:

COG Organization
COG Focus Continuity
and/or
CAP Sustainability

Candidate interactions

Possible future supporting interactions include:

REINFORCES
BUFFERS
PROTECTS

where separately validated.

Critical boundary

If elevated Effort Cost becomes the dominant distinguishing structure,
`PTN-S03 MAINTAINED_WITH_COST` may provide the more specific motif.

Prohibited meanings

This Pattern does not mean:

motivated
excited
determined
energetic personality
high performer

33. PTN-S03 — MAINTAINED_WITH_COST

Class: P2

Status: RESEARCH_ONLY

Canonical meaning

Observable speech-production function remains comparatively maintained while independent Evidence indicates greater acoustic or temporal cost associated with maintaining that output.

Candidate maintained-function signals

One or more:

COG Organization maintained
COG Focus Continuity maintained
REG Stability maintained
CAP Sustainability maintained

plus independently supported:

CAP Effort Cost higher

and/or, after validation:

COG Processing Demand higher

Candidate interactions

Possible supporting relationships:

MASKS
AMPLIFIES
REVEALS

according to the specific upstream geometry.

Preferred scientific concept

maintained function + observable cost

Prohibited meanings

This Pattern does not by itself mean:

burnout
hidden stress
masking emotion
exhaustion
overworking
pushing through
coping failure

34. Maintained function and cost are separate

One of SoulScope's central architectural distinctions is:

What output was maintained?

versus:

What observable cost accompanied it?

Two scans with similar maintained output may differ materially in Effort Cost.

The Pattern layer may summarize that distinction when sufficiently supported.

35. PTN-S04 — CONTEXTUAL_RECONFIGURATION

Class: P1 / P4

Status: RESEARCH_ONLY

Canonical meaning

The vocal pattern changes coherently as the three speaking contexts change while retaining sufficient within-condition organization.

Candidate required structure

After validation:

REG Flexibility higher

plus:

EV_CONTEXT_RECONFIGURATION

and independently supported contextual movement involving one or more of:

COG
CAP
EXP

Coherence requirement

Contextual change alone is insufficient.

Candidate structure should approximately satisfy:

context-sensitive change + coherence – unexplained volatility.

Potential supporting interactions

REDIRECTS
REINFORCES
SHIFTS

where eligible.

Prohibited meanings

It does not mean:

adaptable person
resilient
healthy coping
emotionally flexible
socially skilled

36. PTN-S05 — VARIABLE_RECONFIGURATION

Class: P3 / P4

Status: RESEARCH_ONLY

Canonical meaning

The scan shows substantial contextual or local movement accompanied by comparatively greater variability, volatility, or reduced structural stability.

Candidate structure

Possible support:

REG Stability lower

plus:

substantial DYN volatility
and/or
cross-context reconfiguration

and meaningful movement in another domain such as:

COG
CAP
EXP

Critical distinction

This Pattern is not equivalent to:

low Stability

It requires meaningful multi-domain movement.

Prohibited meanings

It does not mean:

emotionally volatile
unstable person
dysregulated
erratic
unpredictable

37. Contextual versus Variable Reconfiguration

The distinction is conceptualized as:

PTN-S04 = change with greater coherence

and:

PTN-S05 = change with greater volatility.

Neither is intrinsically better or worse.

Exact separating regions require SoulScope calibration.

38. PTN-S06 — CROSS_DOMAIN_MISMATCH

Class: P3

Status: RESEARCH_ONLY

Canonical meaning

Two or more well-resolved functional domains move or resolve in meaningfully different directions, and that mismatch materially changes the interpretation of the scan.

Candidate examples include:

maintained Organization
+
higher Effort Cost

higher Activation
+
narrower expressive deployment

higher Stability
+
lower contextual Flexibility

after relevant constructs are validated.

Evidence requirement

The differing domains must have independent Evidence chains.

Potential supporting interactions

MASKS
REVEALS
CONSTRAINS
REDIRECTS

depending on the specific relation.

Prohibited meanings

It does not mean:

inner versus outer self
hidden emotion
inauthenticity
suppression
deception
conflict inside the person

39. PTN-S03 versus PTN-S06

`MAINTAINED_WITH_COST` is a specific cost-centered motif.

`CROSS_DOMAIN_MISMATCH` is broader.

When:

maintained function
+
higher Effort Cost

is the central validated structure, the more specific:

PTN-S03 `MAINTAINED_WITH_COST`

should generally outrank the generic mismatch candidate.

This is subject to future calibrated selection policy.

40. PTN-S07 — TEMPORAL_RELATION_SHIFT

Class: P4

Status: RESEARCH_ONLY

Canonical meaning

The relationship among independently resolved Constellations changes materially across speaking contexts or compatible longitudinal observations.

Candidate requirement

Preferably supported by:

INT-SHIFTS

or another explicitly validated temporal relation.

At least two distinguishable relation regimes are required:

$$I(t_1) / I(t_2). =$$

Example structure

P1:
stable / lower-cost relation

P2:
higher Activation / higher Cost relation

P3:
maintained output / broader modulation relation

may support a temporal-shift candidate if the relevant upstream constructs are valid.

Prohibited meanings

It does not automatically mean:

recovered
improved
got worse
became positive
resolved the issue
emotion changed

41. Initial Pattern adjacency

The following relationships are provisional research adjacency hypotheses:

PTN-S01 COHERENT_MAINTENANCE
<->
PTN-S02 MOBILIZED_MAINTENANCE

PTN-S02 MOBILIZED_MAINTENANCE
<->
PTN-S03 MAINTAINED_WITH_COST

PTN-S04 CONTEXTUAL_RECONFIGURATION
<->
PTN-S05 VARIABLE_RECONFIGURATION

PTN-S03 MAINTAINED_WITH_COST
<->
PTN-S06 CROSS_DOMAIN_MISMATCH

PTN-S04 CONTEXTUAL_RECONFIGURATION
<->
PTN-S07 TEMPORAL_RELATION_SHIFT

Adjacency SHALL remain calibration-dependent.

No production blend behavior is scientifically established merely by this provisional graph.

42. Pattern modifiers

The Pattern layer may attach at most one modifier.

Candidate modifier vocabulary:

HIGHER_OBSERVABLE_COST
LOWER_OBSERVABLE_COST

HIGHER_MOBILIZATION

LOWER_MOBILIZATION

STRONGER_CONTEXT_SHIFT
LOWER_CONTEXT_SHIFT

BROADER_MODULATION
NARROWER_MODULATION

A modifier requires an independently resolved upstream source.

43. Modifier abstention

If a modifier's source construct is unresolved:

`modifier = OMITTED`

It does not become its opposite.

For example:

`Effort Cost = UNRESOLVED`

does not permit:

`LOWER_OBSERVABLE_COST .`

Unknown remains unknown.

44. Pattern specificity

When several eligible Patterns overlap, selection should prefer the motif that:

- is more specific;
- has stronger independent support;
- has lower contradiction;
- has adequate confidence;
- adds greater explanatory value;
- depends on fewer speculative constructs.

A generic Pattern SHALL NOT outrank a more specific validated Pattern merely because the generic region is easier to satisfy.

45. Complexity control

Pattern selection SHOULD penalize unnecessary semantic complexity.

Conceptually:

$$Score_P = Fit_P - \lambda Complexity_P.$$

Exact implementation is calibration-dependent.

The principle is canonical:

A complex story is not scientifically preferable merely because it uses more information.

46. Incremental-value requirement

A Pattern earns production value only when it communicates something meaningful beyond listing its component Constellations and Interactions.

Before promoting Pattern P , ask:

Does P

provide reproducible useful compression beyond:

Constellation results + Interactions?

If not:

DO_NOT_PROMOTE_PATTERN

47. New Pattern promotion

A new Pattern may be added only when evidence demonstrates that it is:

1. recurrent;

2. reproducible;
3. distinguishable from existing motifs;
4. scientifically meaningful;
5. sufficiently independent;
6. not adequately represented by an existing Pattern plus modifier;
7. understandable to users;
8. bounded against overinterpretation;
9. worth the validation and maintenance cost.

A catchy name is not a promotion criterion.

48. Pattern deprecation

A Pattern should be deprecated if:

- it cannot be reliably distinguished;
- another Pattern subsumes it;
- Pattern + modifier represents it adequately;
- validation fails;
- reliability is inadequate;
- user comprehension is consistently unsafe;
- upstream scientific definitions change materially.

Historical Pattern results retain their original Pattern ID and version.

49. Pattern desirability

No Pattern has intrinsic positive or negative value.

Patterns SHALL NOT be ranked as:

BEST
WORST
HEALTHY
UNHEALTHY
OPTIMAL
IDEAL
GOOD
BAD

A Pattern describes structure.

It does not grade the person.

50. Pattern persistence

A recurring Pattern across compatible scans may be described as:

RECURRING_PATTERN

only when longitudinal compatibility requirements are satisfied.

It SHALL NOT automatically become:

TRAIT
PERSONALITY
TYPE

Repeated measurement remains repeated measurement.

51. Longitudinal Pattern comparison

For compatible scans:

```
PatternTrajectory  
  
scan_ids[]  
  
pattern_results[]  
  
persistence  
transitions[]  
  
geometry_change[]  
  
reference_version  
  
comparison_confidence
```

Possible outputs include:

```
PATTERN_RECURRED  
PATTERN_CHANGED  
PATTERN_BOUNDARY_MOVED  
NO_STABLE_PATTERN  
COMPARISON_LIMITED
```

52. Pattern transition

A Pattern transition:

$$P_A \rightarrow P_B$$

means the registered whole-scan structure changed.

It does not automatically mean:

```
improvement  
decline  
healing  
regression  
progress  
deterioration
```

53. Version compatibility

Longitudinal Pattern comparison requires compatible:

- protocol version;
- Dimension definitions;
- State definitions;
- Interaction definitions;
- Pattern registry;
- relevant models;
- reference behavior.

When scientific compatibility is insufficient:

```
COMPARISON_LIMITED
```


or:

COMPARISON_PROHIBITED

shall be used.

54. Pattern Decision Record

Every Pattern evaluation SHALL create:

```
PatternDecisionRecord

pattern_decision_id
scan_id

pattern_registry_version

upstream_dimension_decision_ids[]
upstream_state_decision_ids[]
upstream_interaction_decision_ids[]

eligible_patterns[]
rejected_patterns[]

candidate_fit[]
candidate_confidence[]

coverage_summary
independence_summary
contradiction_summary

top_candidate
second_candidate

boundary_evaluation

selected_pattern
selected_modifier

publication_status

primary_reason
secondary_reasons[]
```

```
created_at
```

55. Rejected candidate record

Each rejected Pattern SHALL preserve:

```
pattern_id

fit

requirements_passed[]
requirements_failed[]

supporting_structures[]
contradicting_structures[]

missing_required_structures[]

hard_exclusions[]

rejection_reason
```

The Pattern Engine must know why a candidate lost.

56. Alternative Pattern preservation

Where a viable alternative exists, preserve it.

Example:

```
selected:
  PTN-S03 MAINTAINED_WITH_COST

alternative:
  PTN-S01 COHERENT_MAINTENANCE

reason_alternative_lost:
```

independently elevated Effort Cost materially
changed the dominant whole-scan structure

Alternative storage improves traceability without requiring the user to see every candidate.

57. Pattern trace

Every published Pattern SHALL support reconstruction of:

Pattern ← *Interactions* ← *Constellation States* ← *Dimensions* ← *Evidence* ← *Measurements*.

An authorized trace must be able to answer:

1. What Pattern was selected?
2. Which Constellations contributed?
3. Which Dimensions mattered?
4. Which Interactions mattered?
5. Which upstream structures contradicted it?
6. Which alternatives were considered?
7. Why did the selected Pattern win?
8. What required information was missing?
9. Which modifier was used?
10. Which scientific versions generated the result?

58. No new semantics at Pattern level

Every semantic concept in a Pattern must map to defined upstream structures.

For example:

MAINTAINED

must map to authorized combinations involving constructs such as:

Organization
Focus Continuity
Stability
Sustainability

according to the registered Pattern rule.

COST

must map to:

Effort Cost

and/or another separately validated cost-related construct.

Pattern wording SHALL NOT introduce poetic meaning that lacks a corresponding upstream scientific concept.

59. Pattern and Interaction relationship

Interactions are upstream of Pattern.

The Pattern Engine may summarize published Interactions.

It may not create them.

If an Interaction is:

RESEARCH_ONLY

it cannot silently provide production-level Pattern support.

60. Pattern and State relationship

States are upstream of Pattern.

A Pattern does not:

- rewrite a State;
- convert a Boundary Blend into a State;
- infer a State from Pattern identity;
- fill missing State Dimensions.

Continuous geometry remains authoritative.

61. Pattern and Narrative relationship

Narrative receives the completed Pattern Result.

Narrative may:

- explain it;
- simplify language;
- present supporting Constellations;
- communicate uncertainty.

Narrative may not:

- invent a Pattern;
 - select a Pattern;
 - modify a Pattern;
 - strengthen Pattern confidence;
 - create an identity label.
-

62. No LLM Pattern selection

A generative AI system SHALL NOT infer a Pattern from:

- raw audio;
- transcript;
- Evidence tables;
- Constellation prose;
- State labels;
- user notes.

Pattern selection must be executed by the canonical, versioned scientific backend.

The LLM receives only the completed authorized Pattern Result for translation.

63. Pattern and Resonance Signature

The Pattern has no authority over Resonance Signature geometry.

The two branches remain:

Semantic branch

Measurements → Evidence → Dimensions → States → Interactions → Pattern → Narrative.

Visual branch

Qualified Acoustic Measurements(t) → Resonance Signature.

Pattern information SHALL NOT determine Resonance Signature radial structure, temporal geometry, or visual acoustic deposition.

64. Pattern Engine output

Canonical output:

```
PatternResult

scan_id

outcome_type

selected_pattern
selected_modifier

candidate_patterns[]

pattern_fit
pattern_confidence

supporting_constellations[]
supporting_dimensions[]
supporting_interactions[]

contradicting_structures[]

coverage
independence
uncertainty

alternative_pattern

temporal_scope

publication_status
```

```
resolution_reason  
pattern_decision_record  
registry_versions[]
```

65. Example — Coherent Maintenance

Illustrative only:

```
COG:  
  Continuous Organization  
  
REG:  
  Stable Pattern  
  
CAP:  
  Maintained Output  
  
cross-domain contradiction:  
  low
```

Possible research candidate:

```
PTN-S01 COHERENT_MAINTENANCE
```

Bounded meaning:

Several parts of the vocal pattern remained comparatively organized and maintained during the scan.

Not:

The person is resilient or well-regulated.

66. Example — Mobilized Maintenance

Illustrative:

REG Activation:
higher

CAP Mobilization:
higher

COG Organization:
maintained

CAP Sustainability:
maintained

Possible candidate:

PTN-S02 MOBILIZED_MAINTENANCE

Meaning:

Observable mobilization increased while output remained comparatively maintained.

67. Example — Maintained With Cost

Illustrative:

COG Organization:
maintained

COG Focus Continuity:
maintained

CAP Sustainability:
maintained

CAP Effort Cost:
higher

Possible candidate:

PTN-S03 MAINTAINED_WITH_COST

Meaning:

Maintained vocal output coexisted with greater independently measured acoustic or timing cost.

This does not imply hidden stress.

68. Example — Contextual Reconfiguration

Illustrative:

```
P1 -> P2:
  coordinated change

P2 -> P3:
  coordinated change

REG Stability:
  adequate

REG Flexibility:
  higher after validation
```

Possible candidate:

```
PTN-S04 CONTEXTUAL_RECONFIGURATION
```

Meaning:

The vocal pattern changed coherently as the speaking context changed.

69. Example — Variable Reconfiguration

Illustrative:

```
contextual change:
  high

within-prompt volatility:
  high
```

REG Stability:
lower

multi-family discordance:
moderate

Possible candidate:

PTN-S05 VARIABLE_RECONFIGURATION

Meaning:

The scan showed substantial change accompanied by greater short-term variability.

70. Example — Cross-Domain Mismatch

Illustrative:

REG Activation:
higher

EXP modulation:
narrower

CAP Effort Cost:
moderately higher

Possible candidate:

PTN-S06 CROSS_DOMAIN_MISMATCH

Meaning:

Different functional parts of the vocal pattern moved in materially different directions.

It does not imply an inner/outer split.

71. Example — Temporal Relation Shift

Illustrative:

P1:
stable lower-cost relationship

P2:
higher Activation + higher Cost

P3:
maintained output + broader modulation

Possible candidate:

PTN-S07 TEMPORAL_RELATION_SHIFT

Meaning:

The relationship among the measured functional structures changed across the three speaking contexts.

It does not imply Recovery.

72. Example — No Pattern

Suppose:

COG:
Boundary Blend

REG:
Stable

CAP:
Effort Cost unresolved

EXP:
Narrow Range

Interactions:
several low-confidence candidates

Correct result:

NO_PATTERN_PUBLISHED

Reason:

Individual results were informative but did not form one sufficiently supported whole-scan motif.

This is a scientifically valid result.

73. Pattern research strategy

Initial implementation SHOULD prioritize transparent candidate-rule evaluation.

Preferred early architecture:

resolved geometry + States + Interactions + uncertainty + independence → Pattern candidate.

Do not begin with an opaque model that directly assigns Pattern labels.

74. Pattern discovery

As SoulScope accumulates validated data, research may examine recurring whole-scan motifs using methods such as:

- hierarchical models;
- robust clustering;
- latent-profile analysis;
- graph motif analysis;
- mixture models;
- temporal trajectory clustering.

Discovery output SHALL be:

PATTERN_CANDIDATE

not automatically:

PRODUCTION_ACTIVE_PATTERN.

75. Pattern validation sequence

Pattern activation should follow:

Candidate → Replication → Reliability → Incremental Value → Construct Review → Calibration → Generalization

76. User-comprehension requirement

Before a Pattern becomes user-facing, testing SHALL determine whether people incorrectly interpret it as:

- personality;
- diagnosis;
- identity;
- emotional truth;
- positive or negative grade;
- causal explanation;
- stable trait.

If comprehension remains unsafe, the Pattern may:

- remain internal;
 - receive safer user-facing language;
 - or fail production activation.
-

77. Pattern versioning

A MAJOR Pattern Registry change is required when:

- Pattern meaning changes;
- required upstream constructs change;
- Pattern-region logic materially changes;
- adjacency materially changes;
- exclusions change;
- scientific interpretation changes.

Historical Pattern results retain:

```
pattern_id
pattern_version
pattern_model_version
upstream_registry_versions
```

78. Historical Pattern integrity

A new Pattern version SHALL NOT silently rewrite historical results.

If reprocessing is explicitly authorized:

```
original_pattern_result
reprocessed_pattern_result
```

must both remain identifiable.

79. Acceptance tests

PTN-FORCE-001

Given:

```
all Pattern candidates below publication requirements
```

Expected:

```
NO_PATTERN_PUBLISHED
```

Never force the highest candidate.

PTN-UPSTREAM-001

Given:

required upstream Dimension = UNRESOLVED

Expected:

candidate Pattern = INELIGIBLE

PTN-D3-001

Given a Pattern rule requiring:

Recovery
Reserve
or Relational Availability

under the current protocol:

Expected:

INELIGIBLE

PTN-COST-001

Given:

maintained functional output
+
independently elevated Effort Cost

Expected:

PTN-S03 MAINTAINED_WITH_COST may become an eligible research candidate according to active rule status.

It must not be treated as validated merely because it is structurally eligible.

PTN-MULTI-001

Given three materially plausible Patterns:

Expected:

```
NO_PATTERN_PUBLISHED  
  
reason =  
MULTIPLE_PLAUSIBLE_MOTIFS
```

PTN-BLEND-001

Given two eligible adjacent candidates with overlapping support:

Expected:

```
BOUNDARY_PATTERN
```

only if the active boundary policy permits it.

PTN-NONADJ-001

Given two strongly supported nonadjacent candidates:

Expected:

```
NO_PATTERN_PUBLISHED
```

not arbitrary blend.

PTN-MOD-001

Given:


```
one selected Pattern
+
one independently supported permitted modifier
```

Expected:

```
PATTERN_WITH_MODIFIER
```

No new Pattern ID is created.

PTN-MOD-002

Given:

```
modifier source = UNRESOLVED
```

Expected:

```
modifier omitted
```

The opposite modifier SHALL NOT be inferred.

PTN-STATUS-001

Given:

```
required upstream construct = RESEARCH_ONLY
```

Expected:

Pattern cannot become:

```
PRODUCTION_ACTIVE
```

through downstream promotion.

PTN-LLM-001

Given a generative model attempts to select or invent a Pattern:

Expected:

ARCHITECTURE_VIOLATION

PTN-IDENTITY-001

Given user-facing Pattern copy:

You are a Coherent Maintainer.

Expected:

REJECT

PTN-RES-001

Given Pattern selected:

Expected:

Resonance Signature rendering inputs remain unchanged.

No Pattern value enters the time-resolved acoustic renderer.

80. Hard invariants

PTN-INV-01

Pattern reasoning occurs only after completed within-Constellation and Interaction reasoning.

PTN-INV-02

Pattern does not create or change upstream scientific objects.

PTN-INV-03

Pattern is optional.

PTN-INV-04

No Pattern is a personality or identity label.

PTN-INV-05

No Pattern may infer protocol-unobservable constructs.

PTN-INV-06

No candidate normalization may force a winner.

PTN-INV-07

Boundary Patterns require explicit adjacency.

PTN-INV-08

More than two materially plausible candidates normally produce no Pattern.

PTN-INV-09

A Pattern may have at most one Pattern-level modifier.

PTN-INV-10

Missing modifier evidence remains missing.

PTN-INV-11

Pattern scientific status cannot exceed its weakest required upstream construct.

PTN-INV-12

Pattern confidence must account for source-Evidence independence.

PTN-INV-13

A new Pattern requires incremental explanatory value.

PTN-INV-14

Repeated Pattern occurrence does not automatically become a trait.

PTN-INV-15

No Pattern is intrinsically good, bad, healthy, or unhealthy.

PTN-INV-16

Generative AI cannot select or invent Patterns.

PTN-INV-17

Pattern cannot control Resonance Signature geometry.

PTN-INV-18

Historical Pattern results retain version provenance.

PTN-INV-19

Narrative may translate a Pattern but cannot strengthen it.

PTN-INV-20

When no Pattern is adequately supported, no Pattern is the scientifically correct output.

81. What v0.1 freezes

The following are canonical upon adoption:

- Pattern occurs after Interactions;
- Pattern is optional;
- Pattern is a whole-scan motif;
- Pattern describes the scan, not the person;
- continuous upstream geometry remains authoritative;
- initial compact seven-Pattern research grammar;
- one Pattern-level modifier maximum;
- no exhaustive combinatorial naming;
- candidate evaluation does not force a winner;
- explicit no-Pattern outcome;
- explicit Boundary Pattern behavior;
- explicit Pattern adjacency;
- multi-candidate ambiguity may produce no Pattern;

- D3 constructs cannot be recovered through Pattern reasoning;
 - Pattern confidence and Pattern fit remain distinct;
 - Evidence independence propagates into Pattern confidence;
 - all candidate alternatives and rejections remain traceable;
 - LLMs cannot select or invent Patterns;
 - Pattern and Resonance Signature remain scientifically independent;
 - Pattern decisions are versioned;
 - historical results are immutable.
-

82. What v0.1 intentionally does not freeze

The following remain empirical and require SoulScope data:

- exact Pattern-region mathematics;
- Pattern centroids or prototypes;
- covariance or density structures;
- minimum fit thresholds;
- publication thresholds;
- confidence thresholds;
- candidate-separation thresholds;
- Boundary Pattern margins;
- modifier thresholds;
- complexity penalties;
- final adjacency validation;
- exact Pattern weighting;
- whether all seven candidate Patterns survive validation;
- final user-facing Pattern language;
- longitudinal persistence thresholds.

These SHALL NOT be fabricated for implementation convenience.

83. Machine-readable registry recommendation

The executable registry SHOULD represent each Pattern using one canonical definition.

Conceptually:

```
patterns:
- pattern_id
  pattern_version
  canonical_internal_name
  scientific_class
```

```

scientific_status

required_constellations
required_dimensions
required_interactions

optional_dimensions
optional_interactions

required_structural_conditions
hard_exclusions

adjacent_patterns
permitted_modifiers

publication_policy
prohibited_meanings

user_language_key

modifiers:
- modifier_id
  source_construct
  eligibility_policy
  compatible_patterns

global_policy:
  pattern_optional: true
  force_winner: false
  max_modifiers: 1
  allow_unresolved: true
  allow_llm_selection: false

```

Do not duplicate these IDs in multiple unrelated production modules.

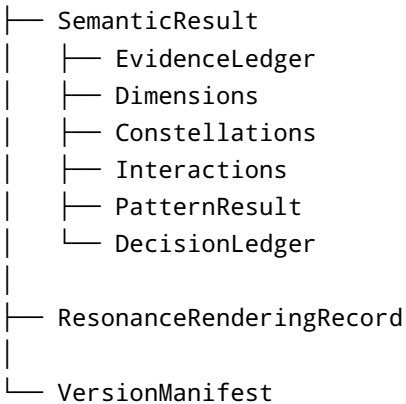
84. Pattern result within Immutable Scan Result

The Pattern result belongs inside:

```

ImmutableScanResult
|
|— AcquisitionRecord
|
|— MeasurementRecord

```



The Pattern is part of the semantic branch only.

85. Final Pattern law

Evidence creates Dimensions

Dimensions create Geometry

Geometry supports States

Completed Constellations support Interactions

Completed reasoning may support a Pattern

but:

A Pattern never creates the reasoning beneath it.

And when the completed reasoning does not support one sufficiently clear whole-scan motif:

NO_PATTERN_PUBLISHED

is not a failure.

It is the correct scientific result.

FINAL PRINCIPLE

The Pattern is the headline of completed evidence, not a new source of evidence.

A Pattern describes what organized this scan. It does not define who the person is.

When the evidence does not support one clean headline, SoulScope does not invent one.